PARTIAL MORTALITY IN THREE SPECIES OF REEF-BUILDING CORALS AND THE RELATION WITH COLONY MORPHOLOGY

Erik H. Meesters, Ineke Wesseling and Rolf P. M. Bak

ABSTRACT

Partial tissue mortality (lesions) in three coral species with distinctly different colony morphologies was quantified in a series of field surveys on a shallow reef. Extent and type of partial mortality was related to differences in size and morphology of the colonies within and between species. Lesion size-frequency distributions were found to be very skewed to the right, meaning that most partial mortality is small in size and well within the regeneration capabilities of all coral species. However, large lesions may make a considerable contribution in terms of surface area to total partial mortality. Within partial mortality two types of lesions must be distinguished. Type I lesions are completely surrounded by living tissue and their occurrence is mostly related to non-bottom associated processes. Type II lesions, at the edge of a colony, are only partly surrounded by living tissue, and are caused mostly by bottom associated processes. The ratio of number of type II/type I lesions differed between species and was related to the ratio of colony edge (circumference)/colony surface area in the three species studied. Type II lesions were larger in size than type I lesions and can make a large contribution to total partial mortality. Species differed in the number of type I and type II lesions per colony and per unit of tissue area. Type II lesions were almost absent on branched colonies, but very frequent on colonies of massive species. Lesion number increased with colony size in the massive species. Most partial mortality is caused by bottom related processes and our results show that "escape in height" is a significant feature in the life history strategies of corals. The surveys showed small colonies to be very vulnerable to partial mortality. Because their circumference/surface area ratio is high, they are very susceptible to colony edge (i.e., bottom-associated) processes that cause mortality. Consequently, small colonies will often suffer whole colony death. On the other hand, large colonies, although unlikely to escape partial mortality, will less often suffer complete mortality. The relationship between this ratio and susceptibility to partial mortality holds as well within species as between species and suggests an important effect of colony morphology on survival. Colony genetic identity also affected susceptibility to partial mortality. Other factors that may influence sensitivity to mortality are discussed. Regeneration capabilities of corals suggest that scleractinian corals have become adapted to the very common occurrence of small lesions.

Damage to the living surfaces of corals is caused by abiotic and biotic disturbances, such as hurricanes (Knowlton et al., 1981; Rogers et al., 1991), diseases (Peters, 1984), emersion at low tide (Fishelson, 1973) predation (Glynn, 1988; Bak and van Eys, 1975; Robertson, 1970) and other interspecific interactions including interactions with humans (Lang, 1971; Pearson, 1981; Bak et al., 1982; Hudson and Diaz, 1988).

Corals and many other taxa are modular colonial organisms and consequently the modules or subunits can function and survive as physiologically independent entities. A group of modules (colony) may be divided into many separate parts through processes such as growth and tissue shrinkage, fission, and fusion (Hughes and Jackson, 1980; Bak et al., 1981; Jackson and Coates, 1986). Partial mortality, the ability of corals to survive the loss of part of the living tissue, is central in their life history strategies. The genetic individual may survive in bits and pieces. Demographic studies of colonial reef organisms taking partial mortality into account suggest that coral colonies loose more tissue through partial

mortality than through whole colony death (Hughes, 1984; Hughes and Jackson, 1985; Done, 1987; Babcock, 1991; Lasker, 1991; McFadden, 1991).

When a colony surface is damaged the tissue around this lesion grows inwards to recover the wound (Fishelson, 1973; Loya, 1976; Bak et al., 1977; Bak and Steward-van Es, 1980; Bak and Criens, 1981; Bak, 1983). If this fails, the result is partial mortality that remains visible as a bare patch of skeleton on the surface of the coral colony.

Regeneration of lesions is influenced by lesion size, interspecific interactions, and environmental conditions. Research has shown that different species are characterized by different regeneration rates, associated with their life history strategies (Bak et al., 1977; Bak and Steward-van Es, 1980; Bak and Criens, 1981; Bak, 1983; Meesters et al., 1992; Meesters and Bak, 1993). If lesions are above a certain size they will not be completely regenerated (Bak and Steward-van Es, 1980) and can be colonized by algae and excavating sponges that may overgrow the coral tissue or weaken the skeleton, respectively.

Obviously, regeneration rates, the occurrence of coral colony surface damage and the role of partial mortality are key elements in the ecology of stony corals. There are no data, however, on the role of these processes on the reef, on possible differences between species and variation between colony growth forms.

The linkage between growth-form/mortality processes is not often considered by reef ecologists, except for factors such as wave action and turbulence (Woodley et al., 1981). Jackson (1979) reviewed morphological strategies in colonial organisms and suggested that the ratio of proximal circumference (colony edge) to colony tissue surface area is an important characteristic in the relative vulnerability of species to bottom-associated processes.

In this study we assess partial mortality (colony surface lesions) on a shallow reef for three species, a branching coral species (*Acropora palmata*) and two massive species (*Porites astreoides* and *Diploria strigosa*). We address the following questions: 1) What are the characteristics of partial mortality for each species? 2) Are lesion characteristics indicative for different causative mechanisms? 3) How do the characteristics of partial mortality relate to different life history strategies?

Materials and Methods

Survey and Species.—Partial mortality was assessed on 446 colonies of 3 abundant coral species. The corals were surveyed at three sites along the coast of the Caribbean island of Curaçao, Netherlands Antilles. The sites were located, in front of the Avila beach hotel, at Carmabi buoy 2, and in front of Seaquarium. At each site colonies were surveyed in 2- or 4-m wide transects depending on density of coral colonies. Transects were parallel to the shore in the shallow reef coral community (Depth 1.5–3.5 m) of the *Acropora palmata* zone and the barren zone (Bak, 1977). We surveyed the most abundant species, notably *A. palmata* (Lamarck, 1816), *Diploria strigosa* (Dana, 1848), and *Porites astreoides* Lamarck, 1816. In each species, colony and lesion characteristics of the first 50 colonies encountered were measured. Due to low densities fewer colonies were surveyed occasionally.

The three species differ in colony morphology and life history traits. *Acropora palmata,* a branching coral, can occur in dense stands that efficiently dissipate the high water energy characteristic of their exposed habitat. Most colonies are well within the breaker zone where they may fragment during storms (Highsmith et al., 1980). Sexual recruits of *A. palmata* were virtually absent on these reefs (Bak and Engel, 1979), and asexual reproduction by fragmentation may constitute the main mode of reproduction (Highsmith, 1982). The two other species also occur in the *A. palmata* zone, but are not exclusive to it, their distribution ranging from 0.5 to 30 m (Bak, 1977). The hemispherical colonies of *Diploria strigosa,* a brain coral, occur in sizes up to 1 m in diameter. *Porites astreoides* colonies are a flat to hemi-spherical, bumpy or smooth polypary surface, not wider than 40 cm. The three species are hermaphroditic (Szmant, 1986). *A. palmata* and *D. strigosa* are broadcast spawners that have one gametogenesis each year and produce few recruits (Bak and Engel, 1979). *P. astreoides* has

a high recruitment success (Bak and Engel, 1979) and releases brooded larvae on an almost monthly basis in Jamaica (Chornesky and Peters, 1987).

Colonies.—A colony was defined as an autonomous mass of skeleton with living tissue. This includes colonies separated by partial mortality into disconnected patches of living tissue but morphologically still one entity. Two patches of living tissue on one colony were recorded as one fission event, three patches as two fission events, etc. The bare skeleton in between was measured and recorded as a lesion. The total surface area of each colony was estimated by regarding it as a circle, a rectangle, a hemisphere, or a combination of a hemisphere and a cylinder, and applying the appropriate geometrical formula. Colony diameter was estimated by taking the square root of the product of the maximal length and perpendicular width.

In *Acropora palmata* three random branches were measured on each colony. Consequently, we only estimated branch surface area. We distinguished upper and undersides of branches. In the two massive species there was no underside. Interspecific comparisons were made without taking branch undersides into account.

Lesion Type and Lesion Size.—A pilot study showed two main types of lesions to exist: a) Type I lesions. These are lesions on all sides surrounded by living tissue; b) Type II lesions. These lesions are only partly surrounded by living tissue. Both type of lesions can be found on the same colony.

Lesion size in the pilot study was found to be approximately log normally distributed. Based on these results and for practical reasons we assigned lesions to one of six size classes: ± 0.1–1 cm^2, 1–4, 4–16, 16–49, 49–100, and larger than 100 cm^2. Lesions larger than 100 cm^2 were measured to the nearest 50 cm^2. To calculate the total lesion area of each colony we used geometric means of the lesion size classes except for lesions larger than 100 cm^2 where the log of the actual size was taken.

RESULTS

Lesion Size-frequency Distributions.—Size and number of type I and II lesions for the three species, for the three sites combined, are shown in the lesion size-frequency distributions of Figure 1. Very obvious is the difference between type I and type II lesions. Distributions of type I lesions, lesions completely surrounded by living tissue, are very skewed to the right, indicating that lesions in the smallest size class dominate the distribution. The distributions of type II lesions, lesions that are only partially surrounded by living tissue, are more evenly spread over the lesion size classes, indicating more lesions in the larger size classes and a higher mean size.

The lesion size-frequency distributions of the three species (Fig. 1) differed depending on species, lesion size and lesion type (test for 3-factor interaction using log-linear model of species, lesion size class, and lesion type; $G = 37.1$, df = 6, $P < 0.001$, the three largest size classes were lumped; Sokal and Rohlf, 1981). The distribution of Type I lesions was the same for *Acropora palmata* (upperside) and *Porites astreoides* ($P = 0.24$), but differed between *Diploria strigosa* and *P. astreoides* ($G = 19.3$, df = 3, $P < 0.001$; three largest size classes lumped). Type II lesions differed between the two non-branching species ($G = 19.9$, df = 5, $P < 0.005$). *Diploria* had fewer lesions in the smallest and more in the largest size class. Type II lesion numbers in *Acropora* were too low for statistical comparisons.

Type I and type II lesions were not equally distributed over the three species ($G = 338.8$, df = 2, $P < 0.00001$). Type II lesions in *Acropora* represented only 7.5% of all lesions counted, while they represented 51 and 69% in *Diploria* and *Porites,* respectively (Fig. 1).

In the branching coral type I lesions were the most common type of lesions (92.5%). Fifty nine percent, of the type I lesions (N = 385), was in the smallest size class. Partial mortality was much more common on the upperside of branches. Of the total number of lesions (N = 416), 93% was located on the upperside of the branches. In contrast to type II lesions, type I lesions never were larger than 100 cm^2.

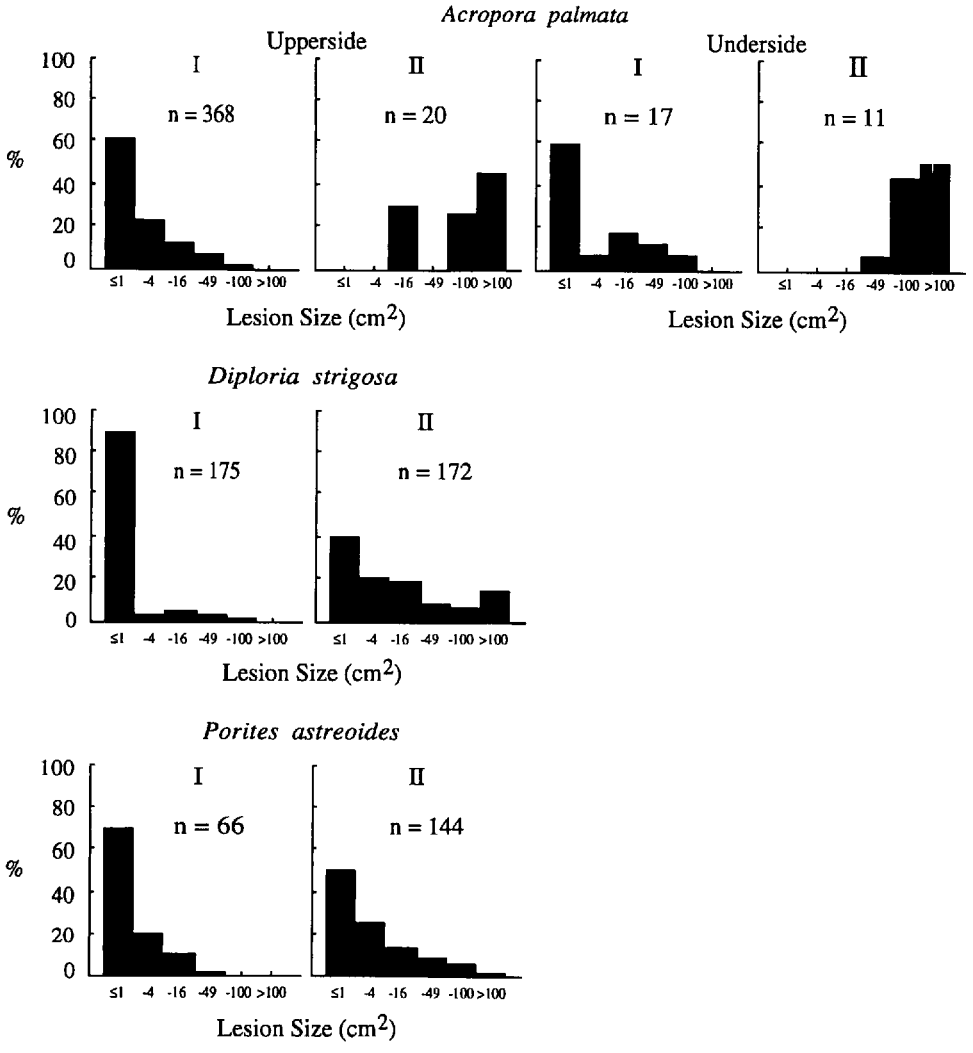

Figure 1. Lesion size-frequency distributions (percentage of total number of lesions) of *Acropora palmata* at upper and underside of branches (N = 333) and for colonies of *Diploria strigosa* (N = 155) and *Porites astreoides* (N = 179). The minimum of each size class is the maximum of the previous size class. Type I lesions are completely surrounded by living tissue; Type II lesions are only partly surrounded by tissue.

In the massive coral species 347 lesions were counted on 155 *Diploria strigosa* colonies, and 210 lesions on 179 *Porites astreoides* colonies. Type II lesions, lesions that are not on all sides surrounded by living tissue, were much more common than in *Acropora*. In *Diploria* 98% of the type I lesions (N = 175) are in the smallest size class.

Lesion Number.—There were large differences in number of lesions between species and lesion types (Table 1). On the average, 68% percent of branches and colonies was damaged. Geometric means most accurately described the means of the highly skewed frequency distributions. *Diploria* has the highest number of lesions per square meter and *Acropora* the lowest. Type II lesions which are

Table 1. Geometric mean number of lesions colony^{-1} and m^{-2} and percent colonies damaged for the three surveyed species. Values for *Acropora palmata* are branch^{-1}. (Roman numbers refer to lesion type; n number of measured colonies)

	I	II	I m^{-2}	II m^{-2}	% Damaged	n
A. palmata	0.215	0.006	1.274	0.011	68	111
D. strigosa	0.061	0.306	0.392	6.355	37	155
P. astreoides	0.017	0.125	0.058	2.467	20	179
Total	0.054	0.096	0.253	1.035	68	445

almost absent on the branching coral, but very abundant, and more frequent than type I lesions, on the two massive species. Remarkable is the low number of type I lesions on *Porites astreoides.* Per unit surface area, type II lesions on *Diploria* and *Porites* are, respectively, 577 and 224 more frequent than on *Acropora.*

The ratio of type II/type I lesions differed also remarkably within each species (Table 1). In *Diploria* type II lesions were five times as frequent as type I lesions, in the other species these ratios are 7.4 and 0.03 for, respectively, *Porites* and *Acropora.*

In the branching coral the mean number of lesions per colony branch depended on the side of the branch and on the type of lesion (Table 2). An examination of the means (Table 3) indicates that on the underside of branches there is no difference between the mean number of type I and type II lesions. On the upperside, Type I lesions are 36 times more frequent than Type II lesions (upperside type I lesions significantly different from all others at $P < 0.01$, other comparisons were not statistically different; multiple comparisons with Tuckey HSD).

Lesion Size.—Mean lesion size (Table 4) differed between species-lesion type combinations (Kruskal-Wallis KW = 246, df = 5, $P < 0.001$). Pairwise comparisons indicated that *Acropora* type II lesions were the largest lesions (Table 5). Within each species type I lesions were always smaller than type II lesions. *Acropora* Type I lesions were larger than *Diploria* and *Porites* type I lesions.

In the branching coral we also measured lesions on the underside of branches (Table 4). Multiple comparisons showed lesions to differ only in size when of different type (Table 6). The side of the branch had no influence on mean lesion size. Using the data on lesion number and lesion size we calculated total lesion area and percentage of the colony area that is lesion area. Partial mortality con-

Table 2. *Acropora palmara.* Relation of lesion abundance to reef location, lesion type, and branch side (3-way anova on the mean number of lesions, type I + type II per colony branch, three branches·colony^{-1}). Data were square-root ($\sqrt{(100 \cdot X)}$ transformed. Branch side, sampling site and lesion type are fixed factors.

Source	SS	Df	MS	F	*P*
Location	35.2	2	17.6	1.08	0.34
Side	1,471	1	1,471	90.54	<0.001
Type	1,367	1	1,367	84.15	<0.001
Side·Location	27.9	2	14.0	0.86	0.42
Side·Type	1,345	1	1,345	82.75	<0.001
Location·Type	27.8	2	13.9	0.86	0.43
Side·Location·Type	36.2	2	18.1	1.11	0.32
Error	7,021	432	16.25		
Total	11,331	443			

Table 3. Number of lesions per branch per colony. (Geometric mean of type I and II lesions of all branches together, for damaged branches, and maximum number per branch)

	Upperside		Underside	
A. palmata	Type I	Type II	Type I	Type II
Mean	0.215	0.006	0.005	0.005
Maximum	6	1	1.67	1
Mean of damaged branches	1.202	0.479	0.41	0.45

stituted less than 0.1% of the total colony area (Fig. 2). Although type II lesions were much larger than type I lesions (Table 5), they were much less frequent (Table 3), especially on the upperside of branches. As a consequence, percent partial mortality of Type I lesions on the upperside was much larger than that in the other three groups (Mann-Whitney *U*-test, $P < 0.001$).

Within the massive species the mean lesion size of type II lesions was larger in *Diploria* than in the smaller *Porites* species, possibly because the upper limits for lesion size are determined by colony size. Because *Diploria* colonies are larger than *Porites* colonies, they can have larger lesions.

Fission.—The number of colonies that actually showed fission, the separation into disconnected parts of living tissue by partial mortality, was low. In *Acropora* fission was absent. In *Diploria* 10 out of 155 colonies (6.4%) showed fission. In these cases fission produced an average of 3.6 colonies per parental colony. In *Porites* fission occurred even less, only in 5 of the 179 colonies (2.8%). Fission in *Porites* produced 2.6 colonies per parent colony.

There was a positive linear relationship between number of colonies produced by fission and colony area in *Diploria* (number of fission derived colonies = 0.76 + 1.86 log (colony surface area); $P = 0.037$, $r^2 = 0.44$). In *Porites,* colony numbers were too low to calculate the effect of colony size on fission. However, the *Porites* colonies with fission were smaller than *Diploria* colonies (geometric mean size respectively 138 and 771 cm^2, $P = 0.027$, 1-way anova) and the fission number of *Porites* (2.6) is very close to the estimated value of 2.2 that can be calculated with the regression equation for *Diploria.* Thus the number of colonies produced by fission, although very low, appears to increase with colony surface area.

Table 4. Partial mortality measured as lesion size (cm^2) in *Acropora palmata, Diploria strigosa,* and *Porites astreoides.* Lesions are grouped into type I (i.e., completely surrounded by tissue) and type II (partly surrounded by living tissue). (In *Acropora palmata* lesions were measured on the upperside (Up.) and underside (Un.) of branches)

Species	Lesion type (side)	Geometric mean	Min.	Max.	Median
A. palmata	I (Up.)	0.91	0.32	70	0.32
	II (Up.)	86.8	8	1,950	70
	I (Un.)	1.23	0.32	70	0.32
	II (Un.)	107	28	400	70
D. strigosa	I	0.45	0.32	70	0.32
	II	3.70	0.32	4,450	2
P. astreoides	I	0.66	0.32	28	0.32
	II	1.42	0.32	200	2

Table 5. Pairwise comparisons of lesion size (Kruskal-Wallis statistic; Siegel and Castellan, 1989). Abbreviations as in Table 1. Species names are given as the first letter of the genus and species name. Significant differences in size: * = $P < 0.05$. Geometric mean size in (cm^2).

	Ap I	Ap II	Ds I	Ds II	Pa I	Pa II
Ap I	0.91	*	*		*	
Ap II	*	86.8	*	*	*	*
Ds I	*	*	0.45	*		*
Ds II		*	*	3.7	*	*
Pa I	*	*		*	0.66	*
Pa II		*	*	*	*	1.42

Inter-colony Differences.—GENETIC EFFECTS.The sampling design used to measure the branching coral can be used to test whether there are differences between colonies in vulnerability. On the average 49.2% of all sampled branches had at least one lesion (Table 7). This proportion did not differ between the three sampling sites ($P = 0.5$, G-test). To determine if colonies differ in their vulnerability to partial mortality we tested whether lesions were randomly distributed over colonies. The expected proportion of branches showing partial mortality when only one branch is sampled of each colony can be calculated as the mean proportion of branches with lesions of all the first, all the second, and all the third branches that were sampled on each colony. With this proportion one can calculate the expected percentage when sampling three branches from the same colony if these are independent of colony identity. This proportion can then be compared with the observed proportion.

When sampling three branches from one colony, the chance that at least one of them is damaged will be $1 - P$ (all three *without* a lesion). For example, for site A this would be $1 - 0.44^3 = 0.92$ [56% with one or more lesions, gives $(1 - 0.56) = 0.44$ with no lesion]. Because *A. palmata* colonies often have many branches, it should be almost impossible for a colony not to have at least one branch with partial mortality.

The expected frequencies and results of the comparison between observed and expected frequencies (Table 7) show that significantly fewer colonies had a lesion than expected when branches were independent of colony identity (paired t-test, $t = 16.2$, df = 2, $P = 0.004$). Test for non-randomness, the runs test (Sokal and Rohlf 1981), indicated that lesions were aggregated within colonies ($P = 0.02$, 0.003, and 0.0001 for sites A, B, and C, respectively). Thus, colonies of *Acropora palmata* differ in their vulnerability to damage.

COLONY SIZE. In the branching coral *Acropora palmata* we investigated the influence of colony size by calculating regression lines of lesion number vs. branch area. The number of lesions increased with total branch surface area, but only a small proportion of the total variation (r^2) was explained by the regression

Table 6. Pairwise comparisons of lesion size in the branching coral *Acropora palmata* (Kruskal-Wallis statistic; overall statistic KW = 94.7, Df = 3, n = 416, $P < 0.001$; Siegel and Castellan, 1989). Abbreviations as in Table 1. Significant differences: * = $P < 0.001$.

	Up. I	Up. II	Un. I	Un. II
Up. I	0.93	*		*
Up. II	*	86.8	*	
Un. I		*	1.23	*
Un. II	*		*	107

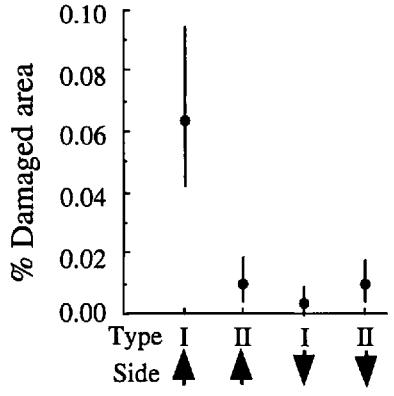

Figure 2. Mean percentage damaged area (95% confidence limits)·colony^{-1} of *Acropora palmata* for type I and II lesions at under (arrow down) and upperside (arrow up) of branches (N = 111 colonies, three branches surveyed colony^{-1}). Type I lesions are completely surrounded by living tissue; Type II lesions are only partly surrounded by tissue. To calculate colony area the upperside of the branch was used.

($P < 0.01$ for all sites, $r^2 = 0.17$, 0.14, and 0.07 for site Seaquarium, Avila, and Buoy 2, respectively). This indicates that branch area is not very important in the number of lesions a colony has.

In both massive species large colonies suffered more from partial mortality, while small colonies often had no partial mortality (Fig. 3). This effect of colony size on the proportion of colonies with/without partial mortality was the same in both massive species (loglinear model $P = 0.38$ for interaction between species and mortality/no-mortality; both species five colony size classes). Average size of undamaged colonies was also significantly smaller than that of damaged colonies (1-way anova, $P < 0.001$ for both species).

The mean number of lesions per colony increased with colony size in both massive species (Fig. 4). The numbers are quite low because there are generally many small colonies (lognormal size distribution) which have almost no lesions. This is reflected in Figure 4 in which most size classes are small. The radius of a hemispherical colony of 75 cm^2 is less than 4 cm. In colonies larger than 75 cm^2, each colony of *Diploria* has on average one lesion; in *Porites* this is one in every two colonies. *Porites* colonies in the smallest size class (average ø = 2 cm)

Table 7. Mean proportion of damaged branches. Means when sampling one branch per colony are calculated from three data groups, the first, second, or third branch from each of the sampled colonies (thus n = 3). Using a binomial distribution the expected proportion when three branches per colony are sampled is calculated. This is then compared with the observed frequency by a paired *t*-test. Sites and expected proportion are compared by a *G*-test. n_{col}, number of colonies each of which three branches were surveyed.

Site	1 branch sampled colony^{-1}	3 branches sampled/colony		
	Mean ± SD	Observed	Expected	n_{col}
Avila	0.56 ± 0.072	0.79	0.92	29
Buoy 1	0.49 ± 0.034	0.71	0.87	34
Seaquarium	0.42 ± 0.073	0.67	0.81	48
Mean ± SD	0.49 ± 0.070	0.82 ± 0.042	0.72 ± 0.061	
Statistical tests	$G = 1.42$ P = 0.49	$G = 1.47$ P = 0.48	$t = 16.2$, Df = 2, P = 0.004	

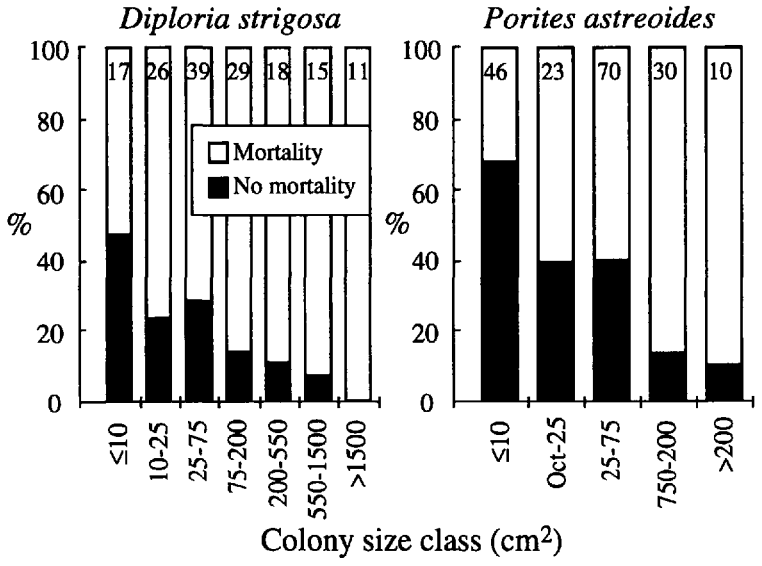

Figure 3. Percentage of colonies with and without partial mortality in colony size classes of *Diploria strigosa* and *Porites astreoides*. Total number of colonies in white bars. Statistics: *Diploria*. $G = 16.9$, df = 6, $P = 0.010$; *Porites*, $G = 26.9$, df = 4, $P < 0.001$.

never had any type I lesions. Small colonies in *Porites* were relatively much more damaged by type II lesions than by type I lesions ($G = 24.8$, df = 4, $P < 0.001$). This effect seems absent in *Diploria*.

Discussion

Our results show clearly that partial mortality differs significantly in different species. These differences are quantitative, e.g., in the number of lesions per

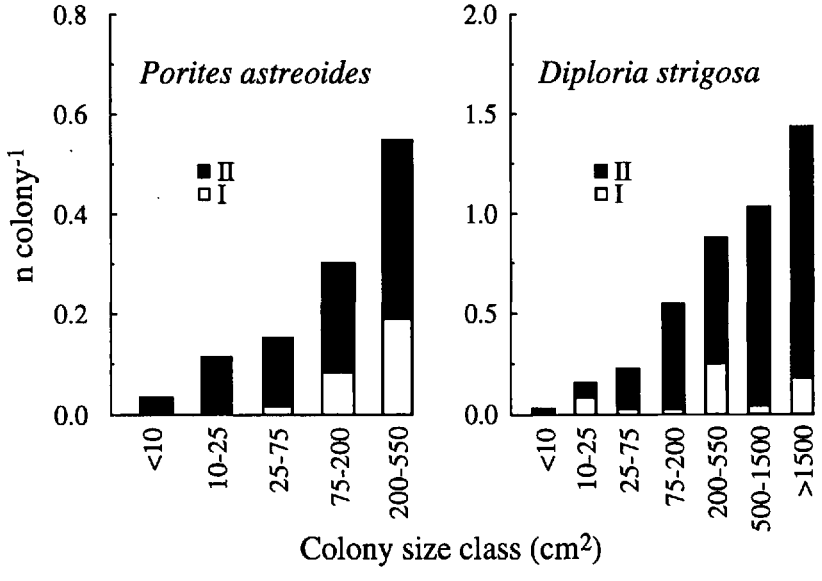

Figure 4. Mean number of type I and type II lesions ·colony^{-1} in colony size classes of *Porites astreoides* and *Diploria strigosa*. Colony numbers as in Figure 3.

colony, and qualitative, e.g., in the occurrence of different types of lesions. Such differences are related to morphological characteristics of the species studied.

Type I versus Type II Lesions.—A remarkable phenomenon in partial mortality is the occurrence and significant differences between two types of lesions: one, surrounded on all sides by tissue (type I), and one which is only partly surrounded by living tissue (type II). Most type I lesions are very small, and are always smaller than type II lesions. As a consequence, a type II lesion will have a stronger impact on a colony than a type I lesion, because the colony will have less tissue to support growth, reproduction, and regeneration. Thus type II lesions are potentially more detrimental to a colony's long-term survival. However, lesion frequencies were highly dependent on species, colony size and lesion type.

The differences between lesion types and species can be explained by differences in the causes of the two lesion types and the effect of colony morphology.

The two types of lesions are probably indicative of different mortality causing processes. Type II lesions are always on the edge of the living tissue (see Materials and Methods) and therefore they occur mostly close to the bottom. Type I lesions are completely surrounded by living tissue and occur over the whole colony, but not on the edge of the living tissue. This means that type II lesions are more likely to originate from agents that are restricted to the bottom, while type I lesions originate more from agents that move freely in the water column. For example, interspecific interactions with corals, sea urchins, and polychaetes, scouring sand, and moving coral fragments are more likely to cause type II lesions. Urchins and polychaetes will start at the edge of the living tissue. Sand and moving fragments are, solely because of gravity, also more likely to cause damage close to the bottom. Fish, divers, and diseases, are likely to cause more lesions of type I.

Colony Morphology.—If we assume that our hypothesis about the origin of the different lesion types is true, *Acropora* will have little chance of type II mortality and most partial mortality should consist of type I lesions. *Porites* has relative to its colony surface area the largest circumference and should have the highest number of type II lesions relative to the number of type I lesions. *Diploria* should be intermediate *Porites* and *Acropora.*

We compared the ratio type II/type I lesions with the ratio of proximal colony circumference/colony surface area for the three species (Fig. 5). The type II/type I ratio not only indicates the relative importance of peripheral versus non-peripheral processes, but to some extent also the total mortality, since type II lesions are larger than type I lesions.

The results (Fig. 5) show that colony morphology is closely related to the specific susceptibility to partial mortality. Branching corals, such as *Acropora,* having a very small proximal-perimeter/surface-area ratio are relatively safe from type II lesions because they have relatively little colony surface close to the bottom. However, because of the small attachment area the risk of breakage is high. In *Acropora* this risk is offset by an "escape in height" from bottom associated processes that cause mortality (Table 1, Fig. 5).

The small attachment area may also explain why type I lesions in the branching coral are larger than in the massive corals. Type I lesions may enlarge through interspecific competition between coral polyps and organisms that settle in the lesions. In massive species the lesion may eventually reach the edge of the tissue and turn into a type II lesion. This is less likely to happen in the branching coral, because of the small attachment area.

Diploria strigosa colonies have much surface area relative to the colony circumference, although less than in the branching coral. Consequently, *Diploria*

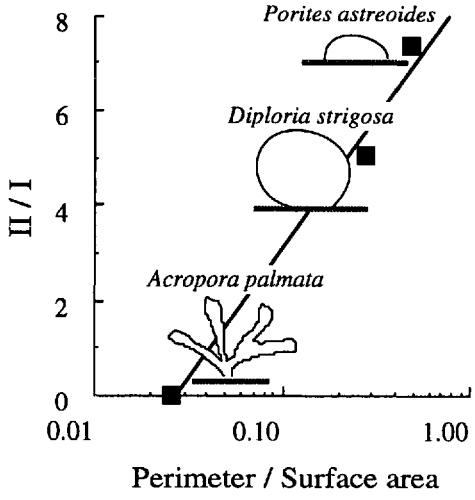

Figure 5. Ratio of the mean number of type I/type II lesions colony^{-1} versus the proximal circumference/total surface-area (cm^{-1}) ratio in *Acropora palmata* (N = 111), *Porites astreoides* (N = 179), and *Diploria strigosa* (N = 155). General colony shape is shown. The perimeter of the proximal end of a *A. palmata* branch was estimated as an ellipse with mean length and width. Perimeter of the other two species was calculated assuming a hemi-spherical morphology for both, using estimated mean colony surface area. Regression: line estimate, II/I = −2.34 + 2.39 ln(100 × circumference/total surface area), r^2 = 0.98, N = 3, P = 0.09.

had more type II partial mortality than *Acropora*. In *Diploria* the lower part of the colony also often overhangs, thus potentially limiting the access for benthic predators.

P. astreoides is more frequently damaged by type II lesions than the other species. Relative to the colony's surface area it has a large perimeter, which renders it relatively more susceptible to damage from benthic organisms or abiotic agents like sand scouring or moving coral fragments.

The low incidence of fission can be explained by the effects of colony morphology on partial mortality. The percentage of colonies originating from fission was much lower than reported earlier (Hughes and Jackson 1985; Babcock 1991). Babcock (1991) concluded that colony fission is likely to constitute a cost rather than a benefit, because there is increased probability of further partial mortality and total fecundity decreases. Colony fission in foliaceous species can be very high (Hughes and Jackson 1985). Colonies of foliaceous species have twice as much perimeter relative to colony surface area as hemispherical colonies of the same diameter (assuming a circular morphology). Consequently, partial mortality in foliaceous colonies will involve more type II lesions than in hemispherical colonies. Because fission involves type II lesions, foliaceous species are more likely to show fission.

Evolutionairy Aspects.—Small lesions are more abundant than larger lesions and most lesions were smaller than 1 cm^2. This size is well within the regeneration capabilities of corals (Bak et al. 1977; Bak and Steward-van Es 1980; Bak 1983; Meesters et al. 1992; Meesters and Bak 1993). Bak and Steward-van Es (1980) noted that regeneration stops after a certain time interval presumably because of an intrinsic factor. Therefore, large lesions, too large to be recovered within the regeneration time interval, will never be closed. The lesion size-frequency distributions do not show the lesions that have been regenerated. Very many small

lesions must recover in a matter of days. The great capacity of corals to regenerate small lesions must have evolved as a consequence of the relatively high frequency with which these lesions are being inflicted. Of course, physiological limitations in the energy translocation necessary for the regeneration are also involved and should be investigated (Meesters et al., 1994).

Colony size influenced the susceptibility to mortality by both massive species. Small colonies often showed no partial mortality, although per unit area they had on average the highest partial mortality. This indicates that small colonies are more likely to escape damage than larger colonies, but when damage occurs, mortality may be total (Hughes and Jackson, 1980, 1985; Rylaarsdam, 1983; Babcock, 1991). Because small colonies have much circumference relative to the total surface area, they will be more susceptible to benthic agents that cause type II lesions. Small colonies of *Porites astreoides* are generally flatter than small colonies of *Diploria strigosa* and this may explain the relative dominance of type II lesions on small *P. astreoides* colonies. Because type II lesions are larger than type I lesions, *P. astreoides* colonies will also more often experience whole colony mortality. The life history strategy of *P. astreoides* allows high juvenile mortality. The species has the second highest rate of recruitment on our reefs, only preceded by *Agaricia agaricites* (Bak and Engel, 1979).

Clearly, the consequences of shape and size are not limited to colonies within a species, but extend to the general colony morphology of any species. Species that have mostly small or flat colonies have much circumference relative to surface area and should be vulnerable to mortality by type II lesions. The vulnerability of species with small maximum size colonies may have led to the development of a set of reproductive traits that is commonly called brooding (Harrison and Wallace, 1990).

In many marine invertebrates brooding is associated with small adult size (Strathmann and Strathmann, 1982). In corals brooding does not appear to be associated with adult polyp size, but with maximum colony size (Kojis and Quinn, 1982; Van Moorsel, 1983; Szmant, 1986; Babcock, 1991; Soong, 1993). Species with small maximum colony sizes generally brood and we hypothesize that this mode of reproduction has evolved to overcome the high mortality that is associated with this type of colony morphology.

Another way to cope with a higher chance of mortality resulting from morphology is to develop a high regeneration potential. This has been shown in platelike colonies of the coral *Meandrina meandrites* (Linnaeus, 1758). This species has the highest regeneration rate ever reported (Meesters and Bak, 1993) and is very successful in interspecific interactions (Lang, 1973). Both characteristics seem to indicate that this species is extraordinarily adapted to the possibility of damage by interactions at its periphery. The absence of partial mortality observed on colonies of the genus *Mycetophyllia* (pers. obs.) is possibly also an indication of highly developed regeneration potential.

Other Factors Affecting Partial Mortality.—Partial mortality is not just related to colony morphology. The results suggest that different genetic characteristics within a species are also important. Branches with partial mortality were more likely to be part of the same colony. This is analogous to the results of Babcock (1991) for massive colonies who found that damaged colonies had more chance of future mortality than undamaged colonies.

The three species may also differ in their susceptibility to disease. The extreme difference in the mean size of type II lesions in *Acropora* when compared with the massive species may be caused by the specific susceptibility of the genus

Acropora to the white band disease (WBD; Bak and Criens, 1981; Gladfelter, 1982). Affected colonies have been recorded continuously ever since this disease has first been reported in the late seventies.

Species may also vary in their attractiveness to predators. *Porites astreoides* appears to be attacked more often by the polychaete *Hermodice carunculata* than *Diploria strigosa* (Ott and Lewis, 1972). Damage by fish bites (Scaridae) has also been observed more often on *P. astreoides* (Meesters, unpubl. data). *Acropora palmata* may be protected by the formation of dense stands and its occurrence in a the normally exposed habitat, both which may limit the accessibility of the colonies to damaging agents.

Another important aspect in partial mortality is regeneration of damage. The three species differ in regeneration potential. *Acropora palmata* has one of the highest regeneration rates found in corals (Bak, 1983; Meesters et al., 1992) and fragments fuse rapidly (Bak and Criens, 1981, 1982). Consequently, fragments have a high survivorship, which enhances asexual recruitment (Highsmith et al., 1980; Bak and Criens, 1981). Regeneration in *Porites* is generally faster than in *Diploria* (Meesters et al., 1992), possibly because of the thicker tissue layer in *P. astreoides*. These differences in regeneration rates may also affect lesion size-frequency distributions.

Conclusions

Our main conclusions are: 1) partial mortality in scleractinian corals is common but mainly small in size and well within the regeneration capabilities of corals; 2) partial mortality is strongly related to size and shape of the colony. Colonies escape in size (less mortality per unit colony area on bigger colonies) and in height (less mortality in branching colonies); 3) The ratio proximal circumference/total surface area of a colony is an indicator of the susceptibility of the colony to tissue mortality. This is NIOZ publication no. 3054.

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ADDRESSES: *(E.H.M. and R.P.M.B.) Institute of Systematics and Population Biology, PO Box 94766, 1090 GT Amsterdam, The Netherlands*; MAILING ADDRESS *(R.P.M.B. and I.W.) Netherlands Institute for Sea Research (NIOZ), PO Box 59, 1790 AB; Den Burg, The Netherlands.*